ARCHITECTURE DOCUMENT

LunaAura: An Explainable Machine Learning Framework for Hormonal and Emotional Health Analysis

1. Introduction

LunaAura is a behavioral analytics platform designed to analyze emotional wellbeing patterns using machine learning and behavioral datasets. The system integrates daily behavioral signals including mood score, perceived stress level, sleep duration, physical activity, and menstrual cycle phase. The goal of the system is to explore how lifestyle behavior and physiological cycles influence emotional wellbeing patterns. The platform provides visual analytics and interpretable machine learning insights rather than diagnosing mental health conditions.

The architecture consists of the following major components:
• Data preprocessing and dataset generation pipeline
• Feature engineering module
• Machine learning model training and evaluation
• Analytics and visualization layer
• Web-based user interface

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2. System Architecture Overview

The LunaAura platform follows a modular web-based architecture where data flows through multiple stages before generating insights for the user.

Architecture Layers

1. Data Source Layer
2. Data Processing Layer
3. Machine Learning Layer
4. Analytics Layer
5. User Interface Layer

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3. High Level Architecture

High-level architecture represents how different system modules interact with each other.

System Flow:

User Interface
↓
Behavioral Data Input
↓
Dataset Processing Pipeline
↓
Feature Engineering Module
↓
Machine Learning Models
↓
Prediction and Analytics
↓
Visualization Dashboard

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4. Data Sources

The system integrates multiple behavioral datasets that represent emotional wellbeing signals.

Primary datasets used:

Depression Dataset

Contains mental health indicators such as:
• PHQ_Total score
• study pressure
• financial pressure
• age
• gender

These variables are used to generate baseline emotional wellbeing indicators.

Sleep and Lifestyle Dataset

Contains behavioral lifestyle variables including:

• sleep duration
• physical activity level

These variables represent daily lifestyle signals affecting emotional wellbeing.

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5. Dataset Generation Pipeline

A master dataset is generated using the script:

build_master_dataset.py

The dataset generation process includes:

1. Loading depression and lifestyle datasets
2. Data cleaning and preprocessing
3. Cycle-aware behavioral simulation
4. Feature engineering
5. Dataset integration

The output dataset is stored as:

master_dataset.csv

This dataset represents daily behavioral observations across multiple users.

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6. Feature Engineering

Several engineered features were created to capture behavioral relationships.

Cycle Features
• cycle_day
• cycle_phase
• luteal_flag

These features represent hormonal cycle stages.

Stress Interaction Features

Stress-cycle interaction was created to capture emotional variation during luteal phase.

stress_cycle_interaction = stress_level * luteal_flag

Sleep-Stress Ratio

Captures buffering effect of sleep.

s*leep_stress_ratio = sleep_duration / (stress_level + 1)*

Hormone-Stress Interaction

Models interaction between hormonal intensity and stress.

h*ormone_stress_interaction = hormone_intensity * stress_level*

Nonlinear Stress Feature

stress_squared = stress_level²

This allows the model to capture nonlinear stress effects.

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7. Machine Learning Architecture

The system trains multiple machine learning models to analyze emotional wellbeing patterns.

Models used:

Logistic Regression

Used as a baseline model for binary classification.

Random Forest

Used for capturing nonlinear relationships between behavioral variables.

Gradient Boosting

Boosting-based ensemble method that improves prediction accuracy by combining multiple weak learners.

Among the evaluated models, Gradient Boosting achieved the highest accuracy.

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8. Model Evaluation

Model performance is evaluated using standard classification metrics:

• Accuracy
• Precision
• Recall
• F1 Score
• ROC Curve
• Confusion Matrix

These evaluation metrics help measure the predictive capability of the model.

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9. Analytics and Visualization Layer

The analytics layer generates visual insights from behavioral datasets and model outputs.

Visualizations include:

• Mood variation across cycle days
• Stress vs mood relationship
• Sleep vs mood relationship
• Cycle phase distribution
• Feature correlation matrix
• Feature importance analysis

These visualizations help identify behavioral patterns affecting emotional wellbeing.

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10. User Interface Architecture

The LunaAura platform provides a web-based user interface for behavioral logging and analytics exploration.

Main UI modules include:

Dashboard

Displays summary metrics and behavioral analytics.

Daily Log

Allows users to input behavioral signals including:

• mood score
• stress level
• sleep duration
• physical activity
• cycle day

Analytics Page

Displays charts and visualizations of behavioral trends.

Model Insights Page

Displays machine learning model insights including feature importance and evaluation metrics.

Research Mode

Provides access to dataset analysis and research insights.

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11. Database Design

The system uses a structured dataset stored in CSV format.

Main dataset:

master_dataset.csv

Important features:

Feature	Description
mood_score	Daily emotional state
stress_level	Perceived stress
cycle_day	Day in menstrual cycle
cycle_phase	Menstrual phase
luteal_flag	Indicator for luteal phase
sleep_duration	Night sleep hours
physical_activity	Activity level
hormone_intensity	Cycle hormone proxy

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12. Data Exchange

Data exchange in the system occurs through the following mechanisms:

Dataset Files

CSV files are used for storing behavioral datasets.

Machine Learning Pipeline

Data is processed using Python scripts and passed to machine learning models.

Visualization Layer

Processed data is displayed through charts using visualization libraries.

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13. Deployment Architecture

The system is designed as a web application consisting of:

Frontend

• HTML
• Tailwind CSS
• Chart.js

Backend

• Python
• Pandas
• Scikit-learn

Data Storage

• CSV datasets

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14. Security and Privacy Considerations

Since the system processes behavioral data, certain privacy considerations must be addressed.

• user identity anonymization
• secure storage of behavioral data
• responsible interpretation of emotional wellbeing predictions

The system is designed as a research platform rather than a clinical diagnostic tool.

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15. Limitations

Current limitations of the architecture include:

• dataset partially generated through simulation
• reliance on self-reported behavioral inputs
• limited historical data for long-term trend analysis
• behavioral charts require consistent daily logging for accurate insights

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16. Future Improvements

Future enhancements include:

• integration with wearable devices
• mobile application deployment
• real-time behavioral data monitoring
• larger real-world datasets for model training
• advanced time-series models for emotional pattern prediction

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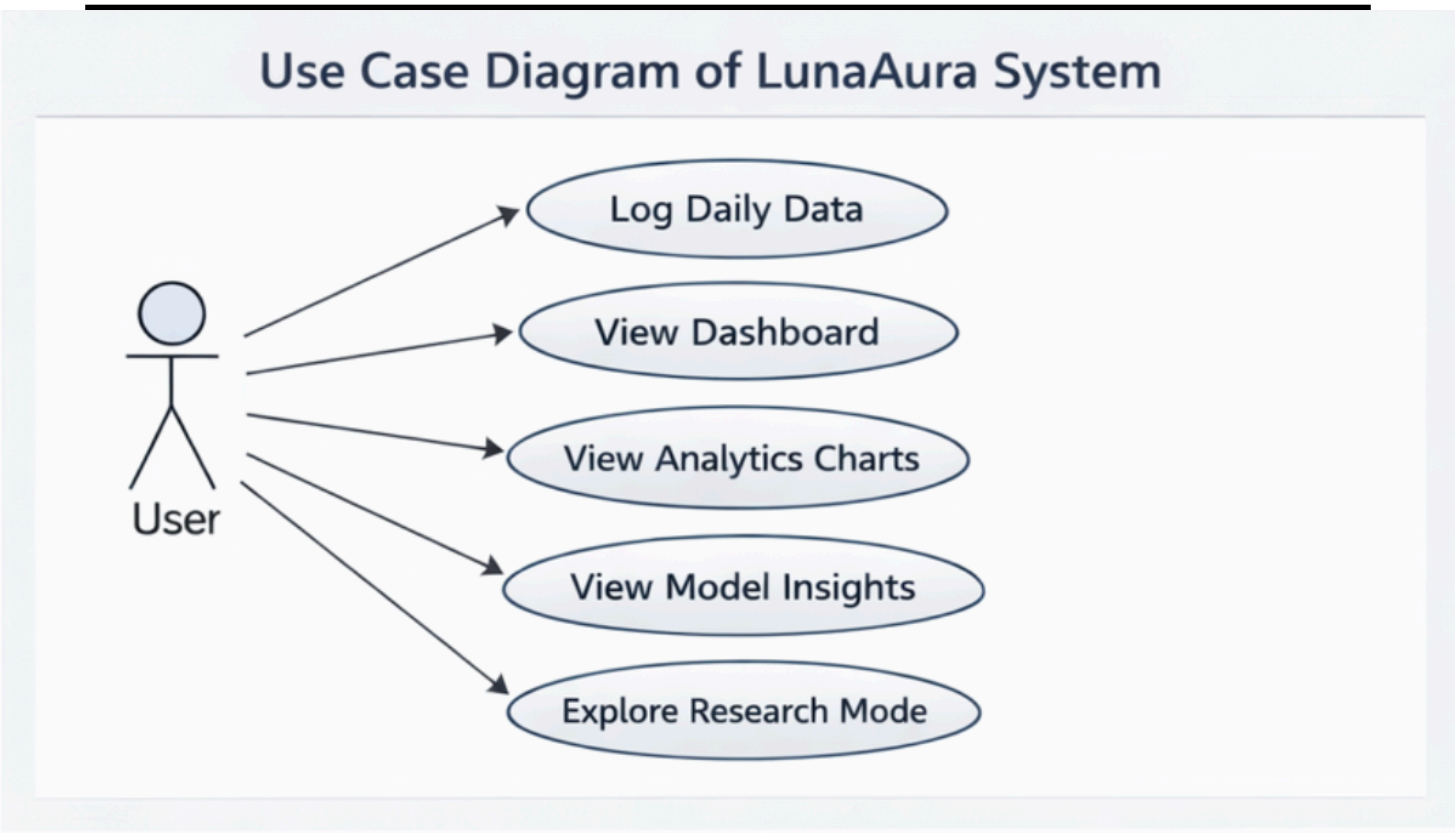
17. Conclusion

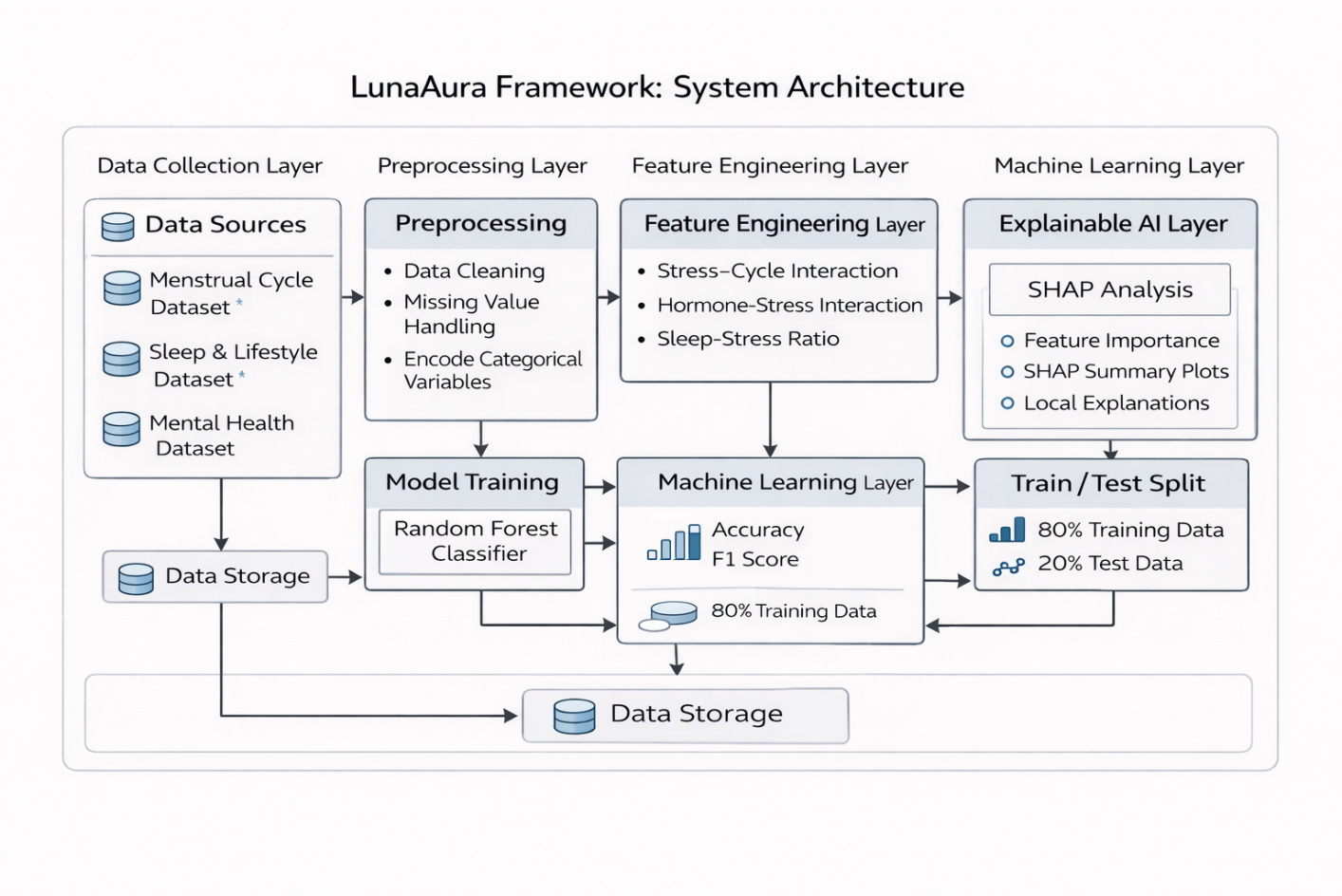
The LunaAura architecture integrates behavioral data processing, machine learning analysis, and visualization modules to explore emotional wellbeing patterns.

The system demonstrates how cycle-aware behavioral signals combined with machine learning can generate meaningful insights into emotional wellbeing trends.

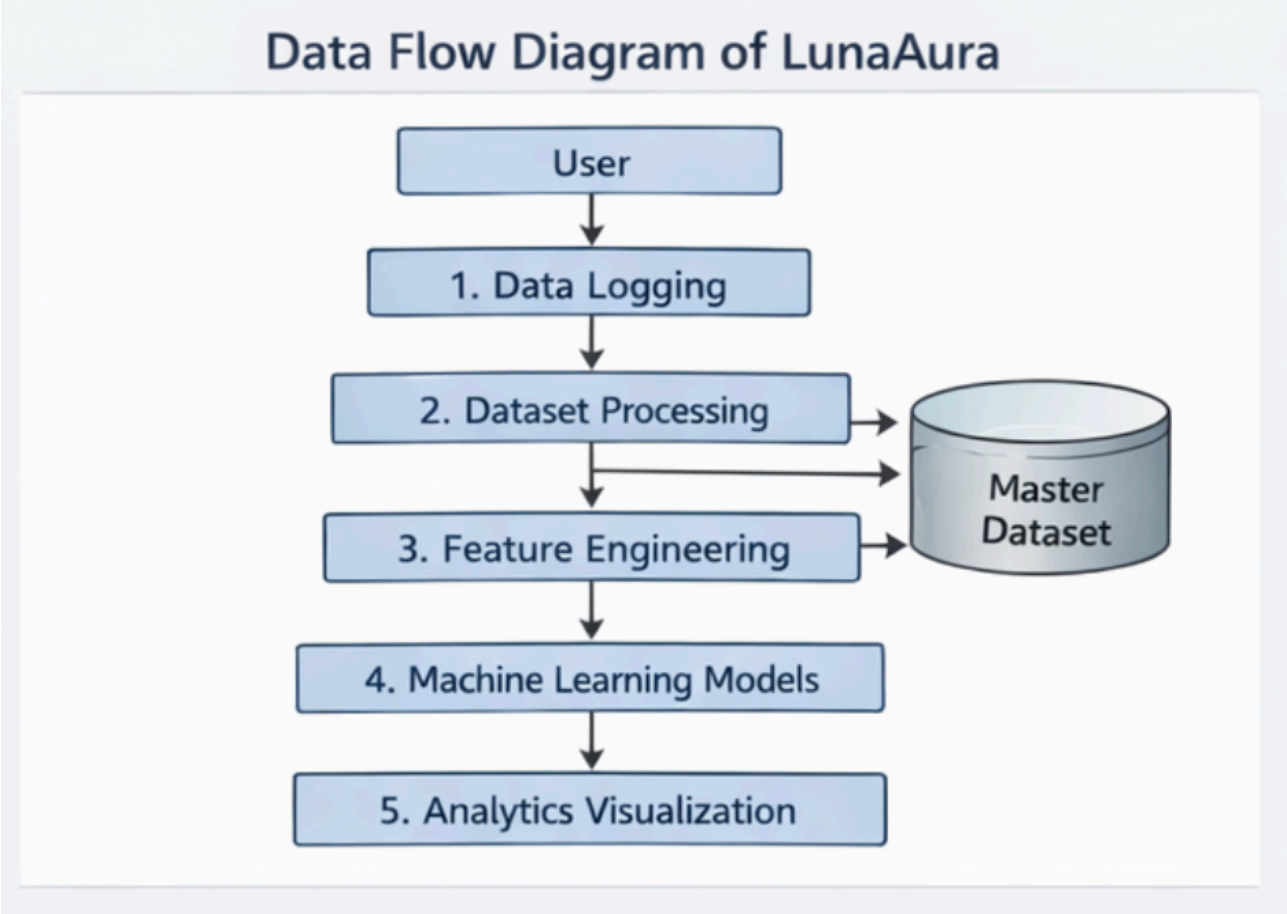
This architecture supports further research in digital mental health analytics and behavioral data analysis.



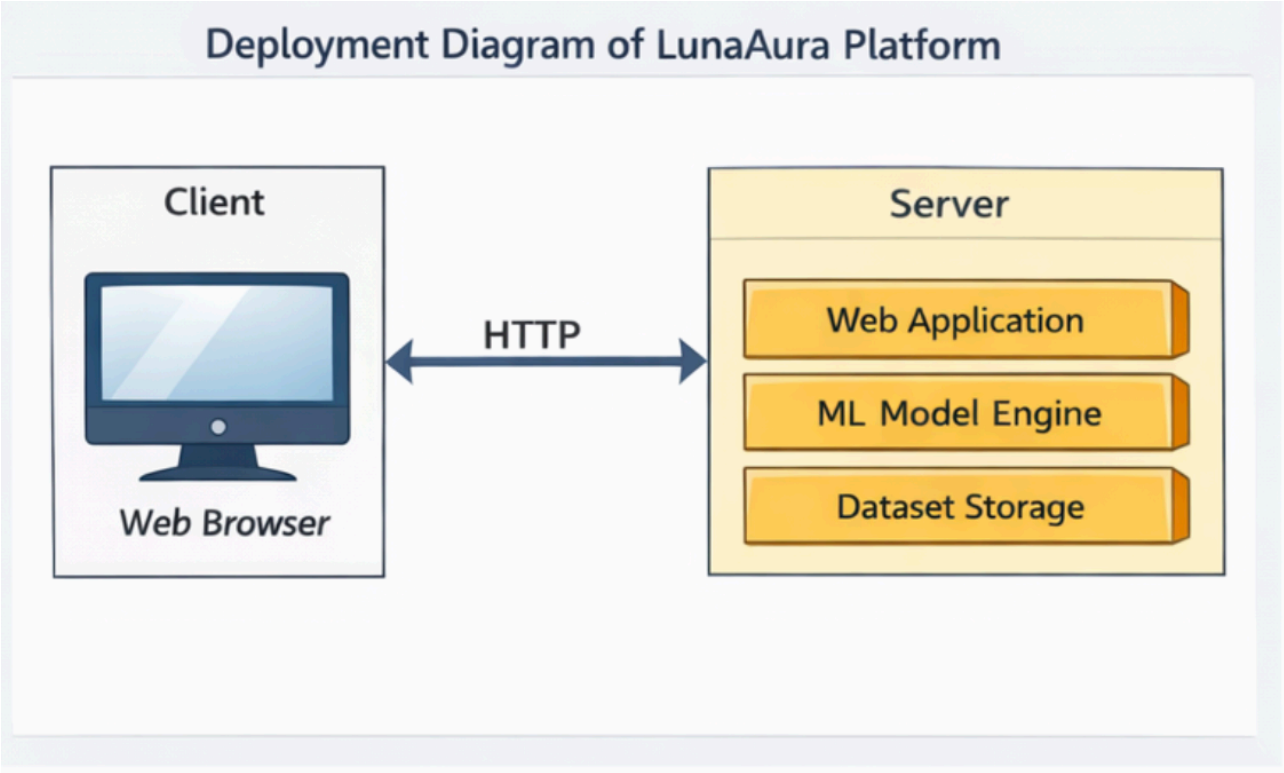
The use case diagram represents the interaction between the user and the LunaAura platform. Users can log daily behavioral data, access the dashboard, analyze behavioral patterns through analytics charts, explore machine learning insights, and access research mode for deeper analysis



The LunaAura architecture processes behavioral and physiological datasets through preprocessing and feature engineering to generate cycle-aware features. A Random Forest model analyzes the processed data using an 80–20 train-test split. SHAP-based explainable AI provides feature importance and interpretable insights into emotional wellbeing patterns influenced by lifestyle and hormonal cycles.



The data flow diagram illustrates how behavioral data moves through the LunaAura system. User inputs are processed through data logging and dataset preprocessing stages. Feature engineering generates interaction variables such as hormone-stress interaction and sleep-stress ratio. The processed dataset is then used by machine learning models to generate predictions, which are visualized through analytics dashboards.



The deployment diagram illustrates how the LunaAura platform operates across client and server components. Users interact with the system through a web browser. The backend server processes behavioral datasets, executes machine learning models, and generates analytics insights which are displayed on the user interface.